

AD23532 – Principles of Data Science

EX:2 :Title: Data Discovery and Exploratory Data Analysis (EDA)

Learning Objectives :

The objective of this experiment is to perform data discovery and exploratory data analysis on a real-world dataset. The experiment helps in understanding the characteristics of the dataset by identifying data types, analyzing statistical summaries, exploring feature distributions, detecting outliers, studying relationships between variables, and discovering hidden patterns before building machine learning models.

Methods Used / Python Libraries Involved and Purpose :

NumPy is used for numerical computations and mathematical operations. It supports statistical calculations and efficient array processing required during exploratory data analysis.

Pandas is used to load the dataset into a DataFrame and perform data exploration tasks such as viewing records, checking data types, finding missing values, generating summary statistics, grouping data, and calculating correlations.

Matplotlib.pyplot is used to create visualizations such as histograms, bar charts, scatter plots, and pie charts. These graphs help understand the distribution of features and identify trends within the dataset.

Seaborn is used for advanced statistical visualizations such as count plots, box plots, pair plots, violin plots, and heatmaps. These plots help analyze relationships between variables, identify outliers, and understand feature interactions.

Describe() Function is used to generate summary statistics including mean, median, minimum, maximum, standard deviation, and quartiles for numerical attributes.

Value Counts is used to determine the frequency distribution of categorical variables such as Gender, Contract Type, Internet Service, and Churn status.

GroupBy() Function is used to compare customer behavior across different categories such as contract type and churn status.

Correlation Analysis is performed to identify positive and negative relationships between numerical variables. This helps in selecting important features for machine learning.

Heatmap is generated using Seaborn to visualize correlation values among numerical features, making it easier to identify highly correlated attributes.

Box Plot is used to detect outliers and compare the distribution of numerical variables across different categories.

Histogram is used to understand the distribution of numerical features such as Tenure, Monthly Charges, and Total Charges.

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PROGRAM CODE:

```
from google.colab import files

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

uploaded = files.upload()

df = pd.read_csv("Customer-Churn.csv")

print("Dataset Shape:", df.shape)

print("\nFirst 5 Rows:")

print(df.head())

print("\nColumn Names:")

print(df.columns.tolist())

print("\nDataset Information:")

df.info()

print("\nStatistical Summary:")

print(df.describe(include='all'))

print("\nMissing Values:")

print(df.isnull().sum())

print("\nDuplicate Rows:")

print(df.duplicated().sum())

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

print("\nMissing Values After Converting TotalCharges:")

print(df.isnull().sum())

df = df.dropna()

if 'customerID' in df.columns:

    df = df.drop('customerID', axis=1)

print("\nCleaned Dataset Shape:", df.shape)

print("\nCustomer Churn Distribution:")

print(df['Churn'].value_counts())

print("\nCustomer Churn Percentage:")
```

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```
print(df['Churn'].value_counts(normalize=True) * 100)

plt.figure(figsize=(6,4))

sns.countplot(data=df, x='Churn')

plt.title('Customer Churn Distribution')

plt.xlabel('Churn')

plt.ylabel('Number of Customers')

plt.show()

print("\nGender Distribution:")

print(df['gender'].value_counts())

plt.figure(figsize=(6,4))

sns.countplot(data=df, x='gender')

plt.title('Gender Distribution')

plt.xlabel('Gender')

plt.ylabel('Number of Customers')

plt.show()

print("\nContract Type Distribution:")

print(df['Contract'].value_counts())

plt.figure(figsize=(8,4))

sns.countplot(data=df, x='Contract')

plt.title('Contract Type Distribution')

plt.xlabel('Contract Type')

plt.ylabel('Number of Customers')

plt.xticks(rotation=15)

plt.show()

print("\nPayment Method Distribution:")

print(df['PaymentMethod'].value_counts())

plt.figure(figsize=(10,4))

sns.countplot(data=df, x='PaymentMethod')

plt.title('Payment Method Distribution')

plt.xlabel('Payment Method')

plt.ylabel('Number of Customers')

plt.xticks(rotation=30)
```

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```
plt.show()

plt.figure(figsize=(7,4))

sns.histplot(data=df, x='MonthlyCharges', kde=True)

plt.title('Monthly Charges Distribution')

plt.xlabel('Monthly Charges')

plt.ylabel('Frequency')

plt.show()

plt.figure(figsize=(7,3))

sns.boxplot(x=df['MonthlyCharges'])

plt.title('Monthly Charges Boxplot')

plt.show()

plt.figure(figsize=(7,4))

sns.histplot(data=df, x='TotalCharges', kde=True)

plt.title('Total Charges Distribution')

plt.xlabel('Total Charges')

plt.ylabel('Frequency')

plt.show()

plt.figure(figsize=(7,3))

sns.boxplot(x=df['TotalCharges'])

plt.title('Total Charges Boxplot')

plt.show()

plt.figure(figsize=(7,4))

sns.histplot(data=df, x='tenure', kde=True)

plt.title('Tenure Distribution')

plt.xlabel('Tenure in Months')

plt.ylabel('Frequency')

plt.show()

plt.figure(figsize=(7,3))

sns.boxplot(x=df['tenure'])

plt.title('Tenure Boxplot')

plt.show()

numerical_features = ['MonthlyCharges', 'TotalCharges', 'tenure']
```

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```
print("\nIQR Outlier Detection:")

for column in numerical_features:

    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df[(df[column] < lower_bound) | (df[column] > upper_bound)]
    print("\nFeature:", column)
    print("Q1:", Q1)
    print("Q3:", Q3)
    print("IQR:", IQR)
    print("Lower Bound:", lower_bound)
    print("Upper Bound:", upper_bound)
    print("Number of Outliers:", len(outliers))

plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Contract', hue='Churn')

plt.title('Contract Type vs Churn')
plt.xlabel('Contract Type')
plt.ylabel('Number of Customers')
plt.xticks(rotation=15)
plt.show()

contract_churn = pd.crosstab(
    df['Contract'],
    df['Churn'],
    normalize='index'
) * 100

print("\nContract Type vs Churn Percentage:")
print(contract_churn)

plt.figure(figsize=(8,5))
sns.countplot(data=df, x='InternetService', hue='Churn')
plt.title('Internet Service vs Churn')
```

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```
plt.xlabel('Internet Service')
plt.ylabel('Number of Customers')
plt.show()

internet_churn = pd.crosstab(
    df['InternetService'],
    df['Churn'],
    normalize='index'
) * 100

print("\nInternet Service vs Churn Percentage:")
print(internet_churn)

plt.figure(figsize=(7,5))
sns.boxplot(data=df, x='Churn', y='MonthlyCharges')
plt.title('Monthly Charges vs Churn')
plt.xlabel('Churn')
plt.ylabel('Monthly Charges')
plt.show()

plt.figure(figsize=(8,5))
sns.histplot(
    data=df,
    x='MonthlyCharges',
    hue='Churn',
    kde=True,
    element='step'
)

plt.title('Monthly Charges Distribution by Churn')
plt.xlabel('Monthly Charges')
plt.ylabel('Frequency')
plt.show()

plt.figure(figsize=(7,5))
sns.boxplot(data=df, x='Churn', y='tenure')
plt.title('Tenure vs Churn')
plt.xlabel('Churn')
```

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```
plt.ylabel('Tenure in Months')
plt.show()

print("\nAverage Tenure by Churn:")
print(df.groupby('Churn')['tenure'].mean())

print("\nMedian Tenure by Churn:")
print(df.groupby('Churn')['tenure'].median())

contract_churn_rate = df.groupby('Contract')['Churn'].apply(
    lambda x: (x == 'Yes').mean() * 100
)

print("\nChurn Rate by Contract:")
print(contract_churn_rate)

print("\nContract Type with Highest Churn:")
print(contract_churn_rate.idxmax())

churned = df[df['Churn'] == 'Yes']

payment_churn = churned['PaymentMethod'].value_counts()

print("\nPayment Methods Used by Churned Customers:")
print(payment_churn)

print("\nMost Frequently Used Payment Method by Churned Customers:")
print(payment_churn.idxmax())

print("\nAverage Tenure by Churn:")
print(df.groupby('Churn')['tenure'].mean())

print("\nAverage Monthly Charges by Contract:")
print(
    df.groupby('Contract')['MonthlyCharges']
    .mean()
    .sort_values(ascending=False)
)

print("\nAverage Monthly Charges by Internet Service:")
print(
    df.groupby('InternetService')['MonthlyCharges']
    .mean()
    .sort_values(ascending=False)
)
```

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)

internet_churn_rate = df.groupby('InternetService')['Churn'].apply(
    lambda x: (x == 'Yes').mean() * 100
)

print("\nChurn Rate by Internet Service:")
print(internet_churn_rate)
print("\nInternet Service with Highest Churn:")
print(internet_churn_rate.idxmax())
print("\nAverage Monthly Charges by Churn:")
print(df.groupby('Churn')['MonthlyCharges'].mean())
print("\nMedian Monthly Charges by Churn:")
print(df.groupby('Churn')['MonthlyCharges'].median())

features = [
    'Contract',
    'InternetService',
    'PaymentMethod',
    'OnlineSecurity',
    'TechSupport',
    'PaperlessBilling'
]

print("\nCategorical Features and Churn Rates:")
for feature in features:
    print("\n", feature)
    result = df.groupby(feature)['Churn'].apply(
        lambda x: (x == 'Yes').mean() * 100
    ).sort_values(ascending=False)
    print(result)

print("\nNumerical Features by Churn:")
print(
    df.groupby('Churn')[
        ['tenure', 'MonthlyCharges', 'TotalCharges']
    ].mean()
```

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```
)
X = df.drop('Churn', axis=1)
y = df['Churn']
y = y.map({'No': 0, 'Yes': 1})
X = pd.get_dummies(X, drop_first=True)
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
print("\nTraining Data Shape:")
print(X_train.shape)
print("\nTesting Data Shape:")
print(X_test.shape)
print("\nTraining Target Shape:")
print(y_train.shape)
print("\nTesting Target Shape:")
print(y_test.shape)
processed_df = X.copy()
processed_df['Churn'] = y
processed_df.to_csv(
    'CustomerChurn_Preprocessed.csv',
    index=False
)
final_df = pd.read_csv('CustomerChurn_Preprocessed.csv')
print("\nFinal Dataset Shape:")
print(final_df.shape)
print("\nFinal Dataset:")
print(final_df.head())
print("\nTotal Missing Values:")
```

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```
print(final_df.isnull().sum().sum())

print("\nFinal File Created Successfully:")

print("CustomerChurn_Preprocessed.csv")

files.download('CustomerChurn_Preprocessed.csv')
```

OUTPUT :

Choose Files Customer-Churn.csv
Customer-Churn.csv(text/csv) - 977501 bytes, last modified: 8/7/2026 - 100% done
Saving Customer-Churn.csv to Customer-Churn.csv
Dataset Shape: (7043, 21)

First 5 Rows:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNWE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	\
0	No phone service	DSL	No	...	No	
1	No	DSL	Yes	...	Yes	
2	No	DSL	Yes	...	No	
3	No phone service	DSL	Yes	...	Yes	
4	No	Fiber optic	No	...	No	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	\
0	No	No	No	Month-to-month	Yes	
1	No	No	No	One year	No	
2	No	No	No	Month-to-month	Yes	
3	Yes	No	No	One year	No	
4	No	No	No	Month-to-month	Yes	

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No

[5 rows x 21 columns]

Column Names:
['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	customerID	7043 non-null	object
1	gender	7043 non-null	object
2	SeniorCitizen	7043 non-null	int64
3	Partner	7043 non-null	object
4	Dependents	7043 non-null	object
5	tenure	7043 non-null	int64
6	PhoneService	7043 non-null	object
7	MultipleLines	7043 non-null	object
8	InternetService	7043 non-null	object
9	OnlineSecurity	7043 non-null	object
10	OnlineBackup	7043 non-null	object
11	DeviceProtection	7043 non-null	object
12	TechSupport	7043 non-null	object
13	StreamingTV	7043 non-null	object
14	StreamingMovies	7043 non-null	object
15	Contract	7043 non-null	object
16	PaperlessBilling	7043 non-null	object
17	PaymentMethod	7043 non-null	object
18	MonthlyCharges	7043 non-null	float64
19	TotalCharges	7043 non-null	object

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Statistical Summary:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	\
count	7043	7043	7043.000000	7043	7043	7043.000000	
unique	7043	2	NaN	2	2	NaN	
top	3186-AJIEK	Male	NaN	No	No	NaN	
freq	1	3555	NaN	3641	4933	NaN	
mean	NaN	NaN	0.162147	NaN	NaN	32.371149	
std	NaN	NaN	0.368612	NaN	NaN	24.559481	
min	NaN	NaN	0.000000	NaN	NaN	0.000000	
25%	NaN	NaN	0.000000	NaN	NaN	9.000000	
50%	NaN	NaN	0.000000	NaN	NaN	29.000000	
75%	NaN	NaN	0.000000	NaN	NaN	55.000000	
max	NaN	NaN	1.000000	NaN	NaN	72.000000	

	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	\
count	7043	7043	7043	7043	...	
unique	2	3	3	3	...	
top	Yes	No	Fiber optic	No	...	
freq	6361	3390	3096	3498	...	
mean	NaN	NaN	NaN	NaN	...	
std	NaN	NaN	NaN	NaN	...	
min	NaN	NaN	NaN	NaN	...	
25%	NaN	NaN	NaN	NaN	...	
50%	NaN	NaN	NaN	NaN	...	
75%	NaN	NaN	NaN	NaN	...	
max	NaN	NaN	NaN	NaN	...	

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	\
count	7043	7043	7043	7043	
unique	3	3	3	3	
top	No	No	No	No	
freq	3095	3473	2810	2785	
mean	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	

	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	\
count	7043	7043	7043	7043.000000	
unique	3	2	4	NaN	
top	Month-to-month	Yes	Electronic check	NaN	
freq	3875	4171	2365	NaN	
mean	NaN	NaN	NaN	64.761692	
std	NaN	NaN	NaN	30.090047	
min	NaN	NaN	NaN	18.250000	
25%	NaN	NaN	NaN	35.500000	
50%	NaN	NaN	NaN	70.350000	
75%	NaN	NaN	NaN	89.850000	
max	NaN	NaN	NaN	118.750000	

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```
TotalCharges Churn
count          7043  7043
unique         6531    2
top            No
freq           11  5174
mean          NaN   NaN
std           NaN   NaN
min           NaN   NaN
25%           NaN   NaN
50%           NaN   NaN
75%           NaN   NaN
max           NaN   NaN
```

```
[11 rows x 21 columns]
```

Missing Values:

```
customerID      0
gender          0
SeniorCitizen   0
Partner         0
Dependents      0
tenure          0
PhoneService    0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
```

Duplicate Rows:
0

Missing Values After Converting TotalCharges:

```
customerID      0
gender          0
SeniorCitizen   0
Partner         0
Dependents      0
tenure          0
PhoneService    0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
Contract        0
PaperlessBilling 0
PaymentMethod   0
MonthlyCharges  0
TotalCharges    11
Churn           0
dtype: int64
```

Cleaned Dataset Shape: (7032, 20)

Cleaned Dataset Shape: (7032, 20)

Customer Churn Distribution:

Churn

No 5163

Yes 1869

Name: count, dtype: int64

Customer Churn Percentage:

Churn

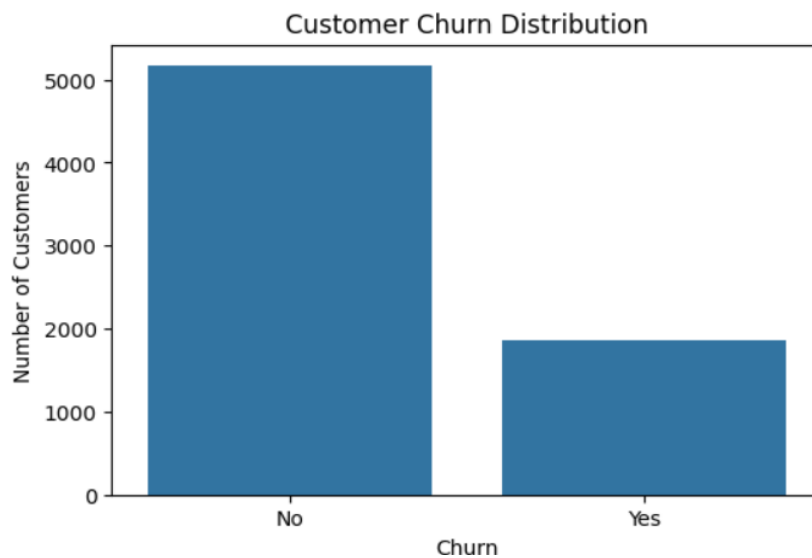
No 73.421502

Yes 26.578498

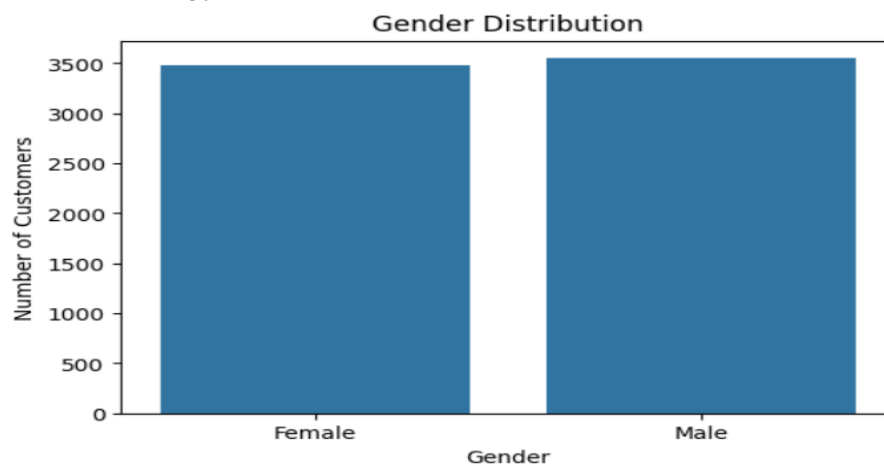
Name: proportion, dtype: float64

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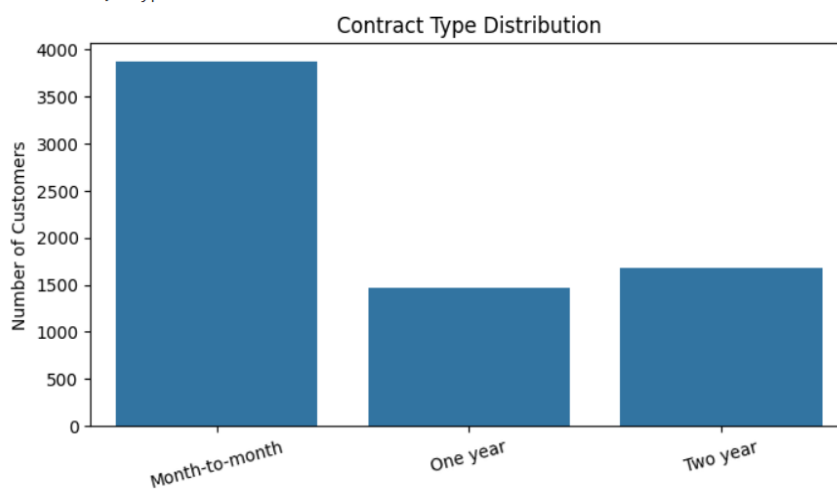
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Gender Distribution:
gender
Male 3549
Female 3483
Name: count, dtype: int64

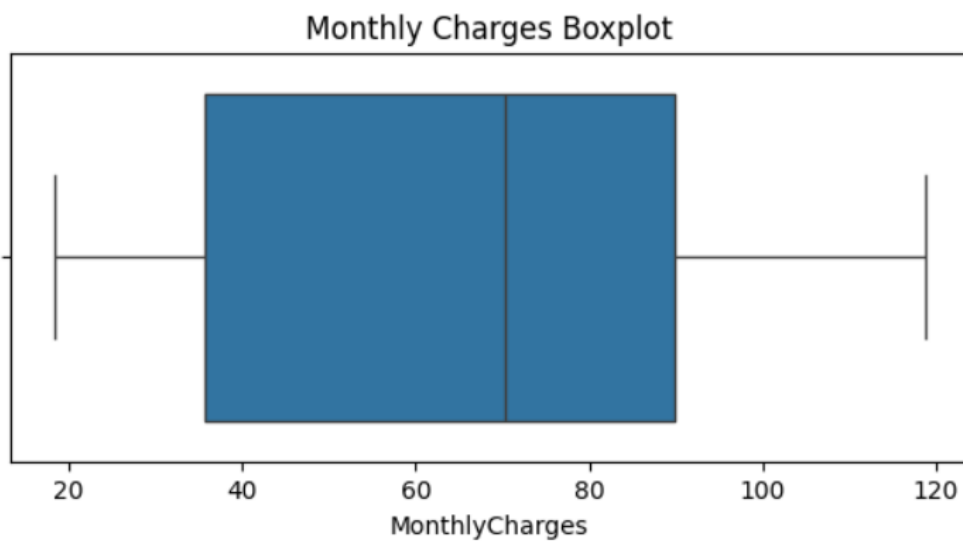
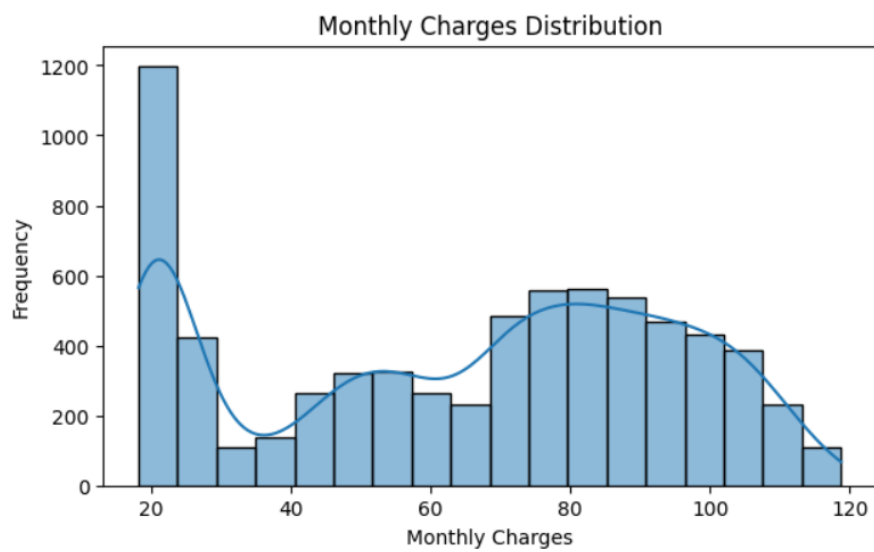
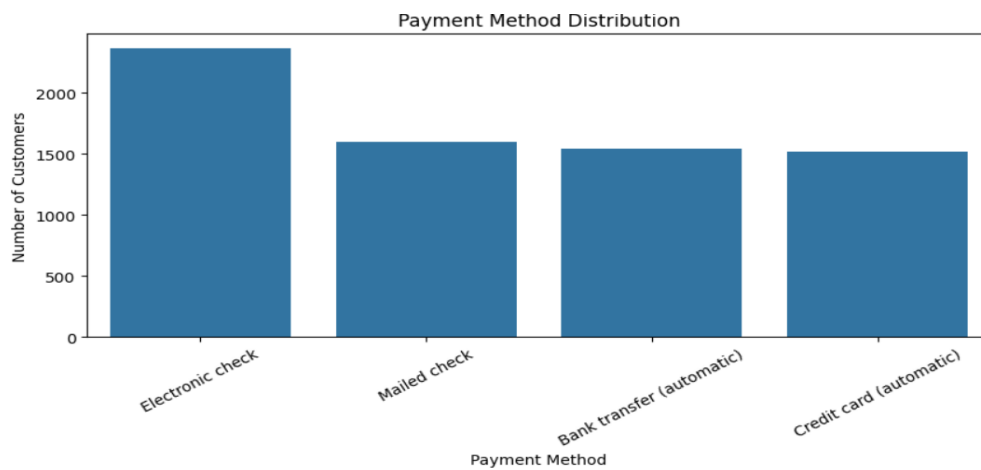


Contract Type Distribution:
Contract
Month-to-month 3875
Two year 1685
One year 1472
Name: count, dtype: int64



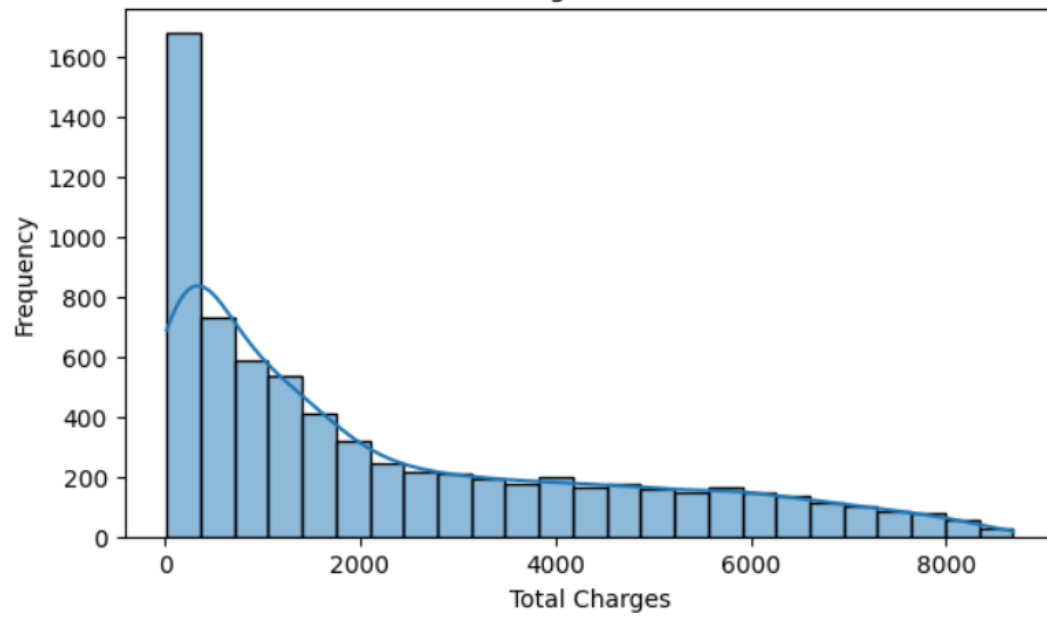
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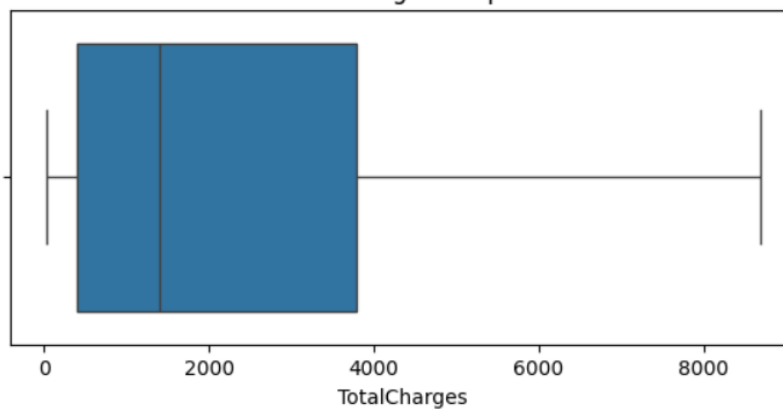


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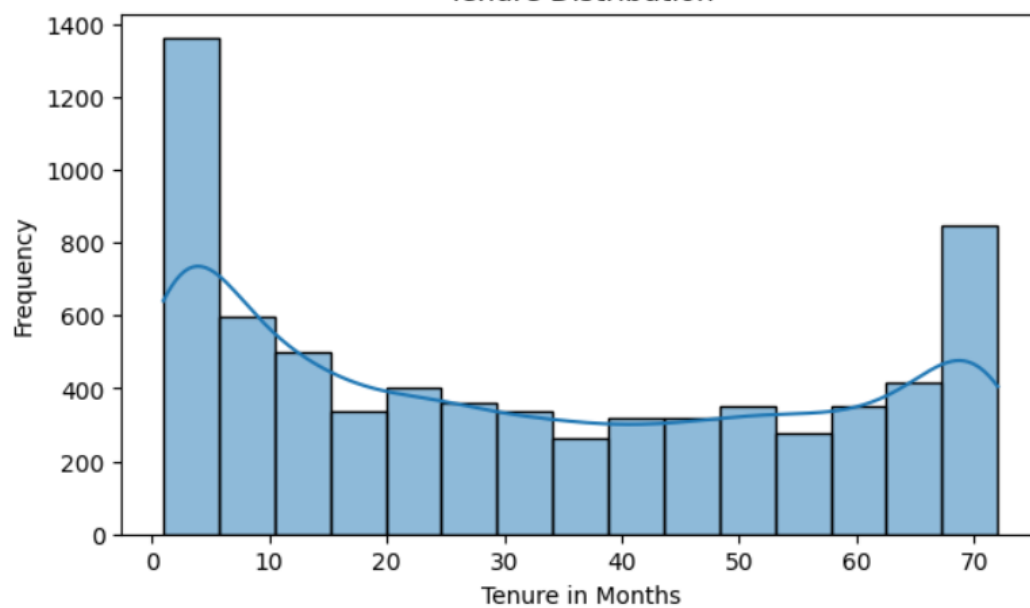
Total Charges Distribution



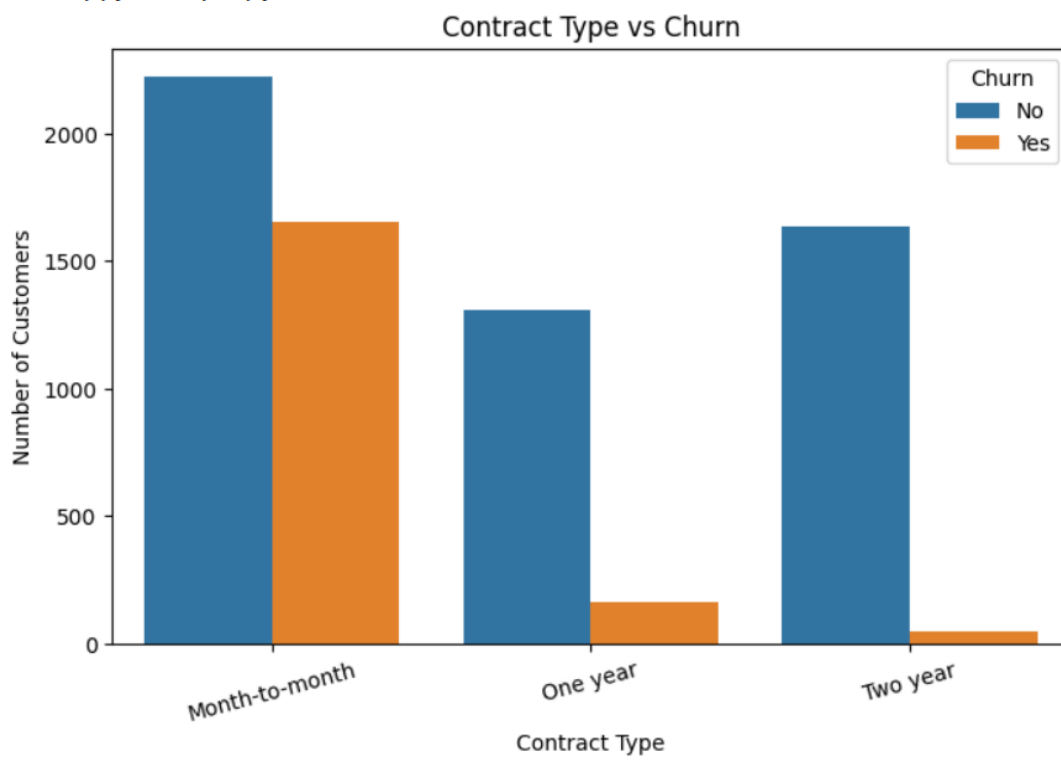
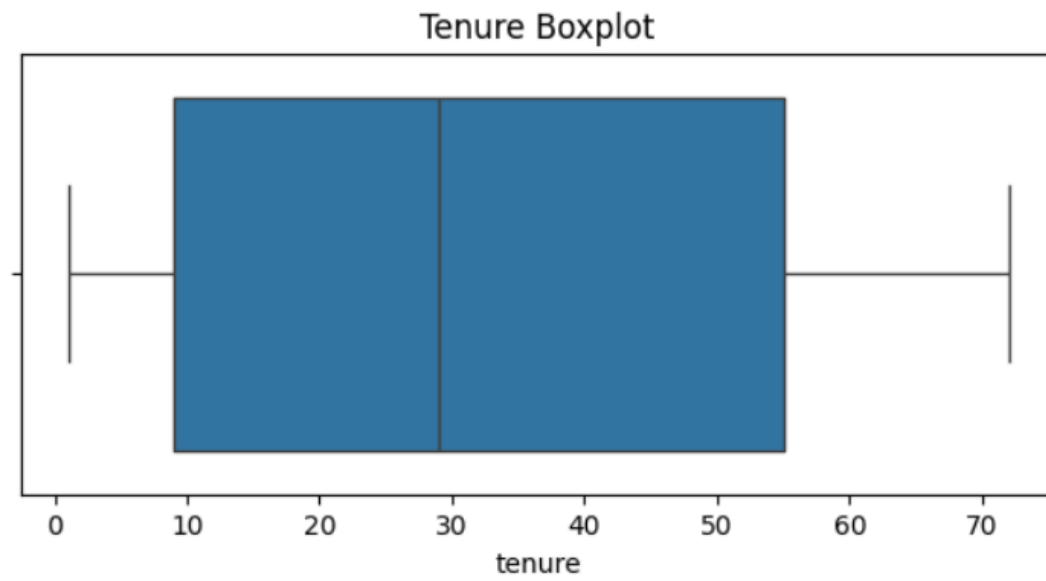
Total Charges Boxplot



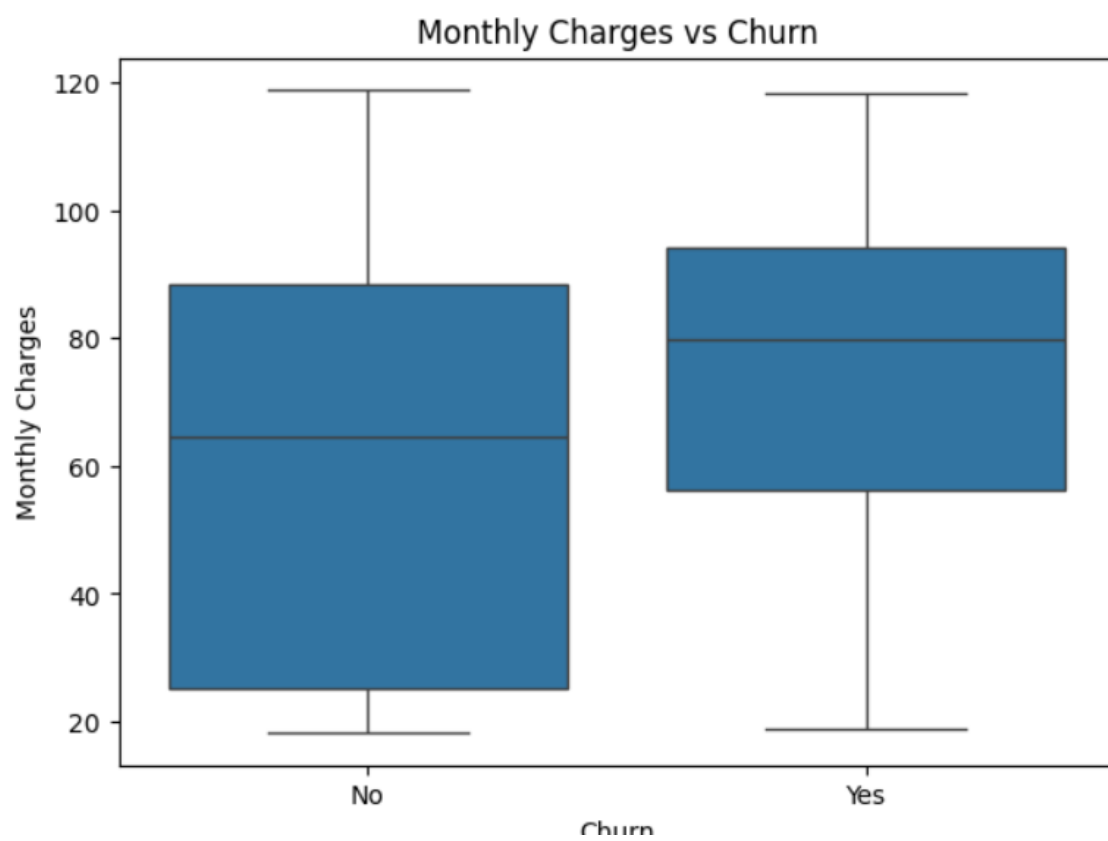
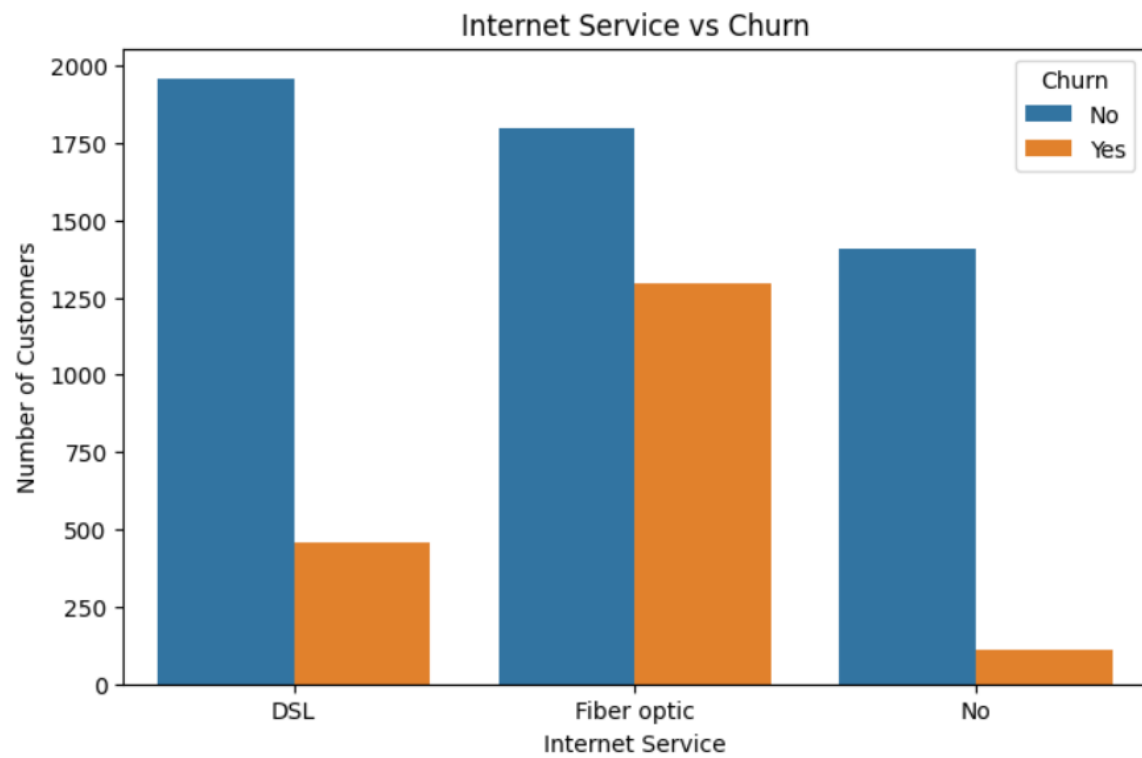
Tenure Distribution



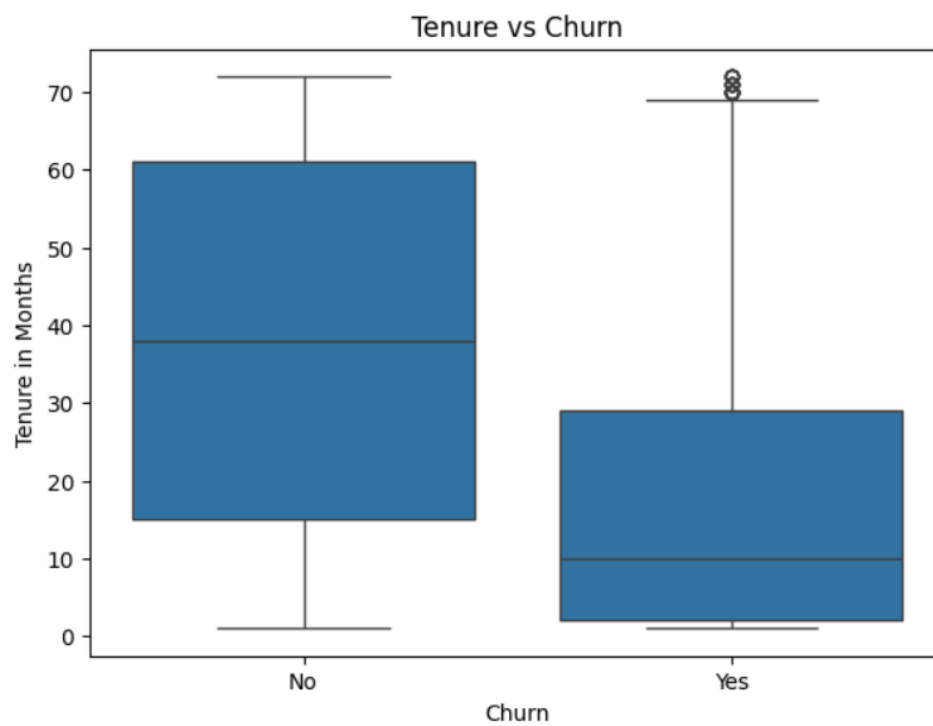
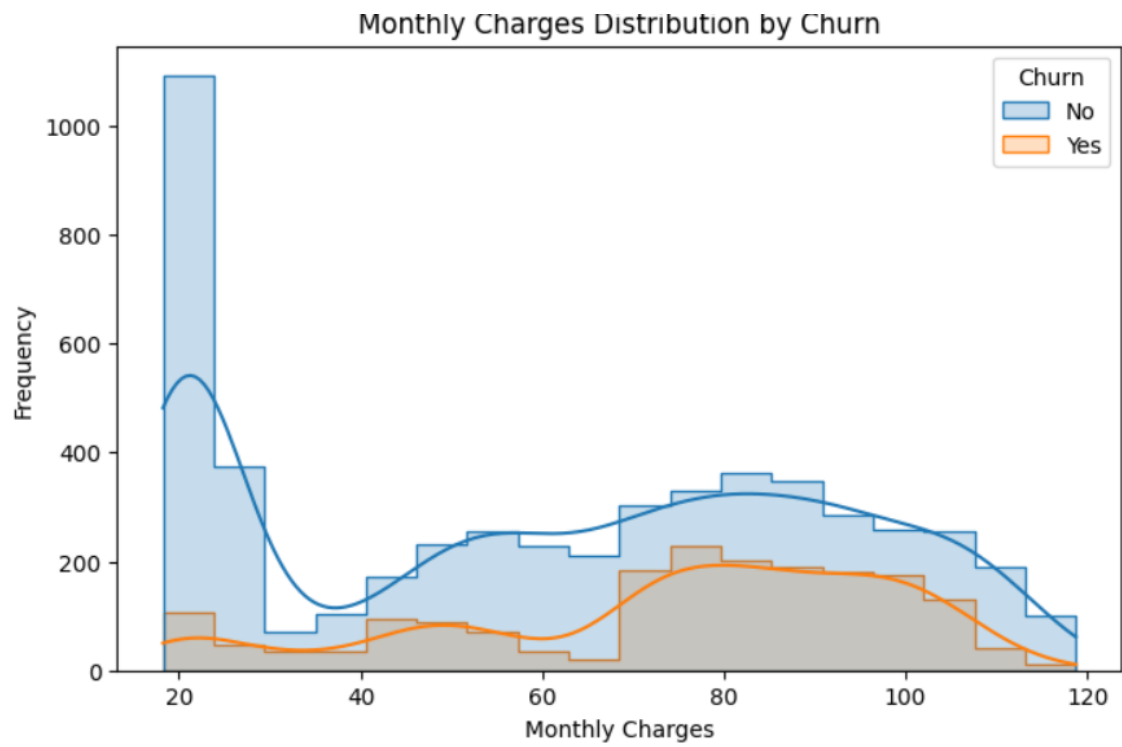
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```
... Final Dataset:
SeniorCitizen tenure MonthlyCharges TotalCharges gender_Male \
0 0 1 29.85 29.85 False
1 0 34 56.95 1889.50 True
2 0 2 53.85 108.15 True
3 0 45 42.30 1840.75 True
4 0 2 70.70 151.65 False

Partner_Yes Dependents_Yes PhoneService_Yes \
0 True False False
1 False False True
2 False False True
3 False False False
4 False False True

MultipleLines_No phone service MultipleLines_Yes ... StreamingTV_Yes \
0 True False ... False
1 False False ... False
2 False False ... False
3 True False ... False
4 False False ... False
```

IQR Outlier Detection:

Feature: MonthlyCharges

Q1: 35.5875

Q3: 89.8625

IQR: 54.275

Lower Bound: -45.824999999999996

Upper Bound: 171.27499999999998

Number of Outliers: 0

Feature: TotalCharges

Q1: 401.45

Q3: 3794.7375

IQR: 3393.2875000000004

Lower Bound: -4688.481250000001

Upper Bound: 8884.66875

Number of Outliers: 0

[5 rows x 31 columns]

Total Missing Values:

0

Final File Created Successfully:

CustomerChurn Preprocessed.csv

Feature: tenure

Q1: 9.0

Q3: 55.0

IQR: 46.0

Lower Bound: -60.0

Upper Bound: 124.0

Number of Outliers: 0

Learning Outcome :

From this experiment, I learned how to perform data discovery and exploratory data analysis using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. I gained practical knowledge in understanding dataset characteristics, analyzing statistical summaries, exploring feature distributions, identifying missing values and outliers, studying relationships between variables, and visualizing important patterns. These insights help in understanding the data more effectively and support better decision-making before developing machine learning models.